Supervised Learning

Supervised learning trains a model on labeled examples. Every training example pairs an input with a known target value, and the learning algorithm searches for a function that maps inputs to targets. The two main supervised tasks are classification, where the target is a category such as spam or not spam, and regression, where the target is a continuous number such as a house price. A model is judged by how well it predicts targets for examples it has never seen. Overfitting happens when a model memorises noise and quirks of the training set, so training error becomes very low while error on new data stays high. Underfitting is the opposite problem: the model is too simple to capture the real pattern.

Unsupervised Learning

Unsupervised learning looks for structure in data that has no labels. Clustering groups similar items together, and K-means is a popular clustering algorithm that repeatedly assigns each point to its nearest centroid and then moves each centroid to the mean of its assigned points. The number of clusters K must be chosen by the practitioner. Dimensionality reduction compresses many features into a few while keeping the most important variation; principal component analysis (PCA) does this by projecting data onto directions of maximum variance. Unlike supervised learning, there is no single correct answer to check against, so evaluation relies on internal measures such as cluster cohesion or on human judgement.

Evaluating Classifiers

Accuracy is the fraction of predictions that are correct, but it can be misleading when classes are imbalanced. Precision is the fraction of predicted positives that are truly positive, and recall is the fraction of true positives that the model finds. The F1 score is the harmonic mean of precision and recall. Cross-validation estimates generalisation by splitting the data into k folds, training on k-1 folds and testing on the remaining fold, then averaging the k results. A held-out test set must never be used for tuning, otherwise the reported score is optimistically biased.

Bias, Variance and Regularization

The bias-variance tradeoff describes two sources of error. High bias means the model makes strong simplifying assumptions and underfits. High variance means the model is very sensitive to the particular training sample and overfits. Regularization reduces variance by penalising model complexity. L2 regularization, also called ridge, adds the sum of squared weights to the loss, while L1 regularization, also called lasso, adds the sum of absolute weights and can drive some weights exactly to zero, which performs feature selection. Early stopping and gathering more training data also reduce overfitting.